

5.2. ASSET ALLOCATION WITH INFREQUENT TRADING

In Ang, Papanikolaou, and Westerfield (2013), I develop an asset allocation model in which the investor can transact illiquid assets only at randomly occurring liquidity events. This notion of illiquidity is that usually illiquid assets are just that—illiquid and cannot be traded. But when the liquidity event arrives, investors can trade.

I model the arrival of liquidity events by a Poisson arrival process with intensity λ . The interval between liquidity events is $1/\lambda$. For real estate or private equity, intervals between trading would occur every ten years or so, so $\lambda = 1/10$. As λ increases to infinity, the opportunities to rebalance become more and more frequent and in the limit approach the standard Merton (1981) model where trading occurs continuously. Thus λ indexes a range of illiquidity outcomes.

Poisson arrival events have been used to model search-based frictions since Peter Diamond (1982), who won his Nobel Prize in 2010. The following year, he was nominated to serve on the Federal Reserve Board of Governors, but Republican opposition blocked his confirmation.

Illiquidity risk causes the investor to behave in a more risk-averse fashion toward both liquid and illiquid assets. Illiquidity risk induces time-varying, endogenous risk aversion. Harvard discovered in 2008 that although it is wealthy, it cannot “eat” illiquid assets. Illiquid wealth and liquid wealth are not the same; agents can only consume liquid wealth. Thus the solvency ratio of illiquid to liquid wealth affects investors’ portfolio decisions and payout rules—it becomes a state variable that drives investors’ effective risk aversion.

The takeaways from the Ang, Papanikolaou and Westerfield model are:

Illiquidity Markedly Reduces Optimal Holdings

Start with the bottom line in Panel A of Table 13.6, which reports a baseline calibration where the investor holds 59% in a risky asset that can always be traded. This weight is close to the standard 60% equity allocation held by many institutions. As we go up the rows, the asset becomes more illiquid. If the risky asset can be traded on average every six months, which is the second to last line, the optimal holding of the illiquid asset contingent on the arrival of the liquidity event is 44%. When the average interval between trades is five years, the optimal allocation is 11%. For ten years, this reduces to 5%. Illiquidity risk has a huge effect on portfolio choice.

Rebalance Illiquid Assets to Positions Below the Long-Run Average Holding

In the presence of infrequent trading, illiquid asset wealth can vary substantially and is right-skewed. Suppose the optimal holding of illiquid assets is 0.2 when the liquidity event arrives. The investor could easily expect illiquid holdings to vary from 0.1 to 0.35, say, during nonrebalancing periods. Because of the right-skew, the average holding of the illiquid asset is 0.25, say, and is greater than the optimal